

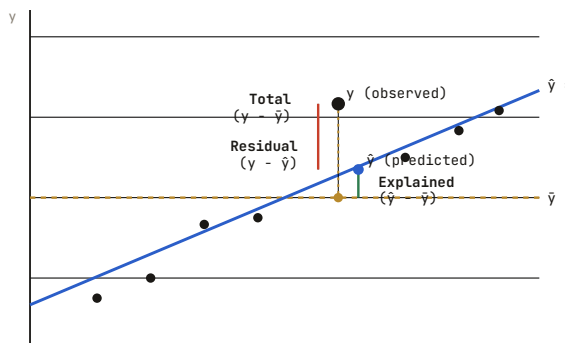
The Anatomy of a *Regression* Line

Six numbers do the heavy lifting when you fit a line to data. Here is what each one actually *looks* like — drawn, decomposed, and demystified.

FOUNDATION

Three kinds of *distance*

Every point in your dataset has three vertical distances that matter. Understanding them is the whole game — SST, SSE, and SSR are just the squared versions, summed up.



Pick any observation y . Its distance from the *mean* $\bar{y}$ is the **total** deviation. The regression line captures part of it — the **explained** part, $\hat{y} - \bar{y}$. Whatever is left between the point and the line is the **residual**, $y - \hat{y}$.

Total (from $\bar{y}$)

Explained (by line)

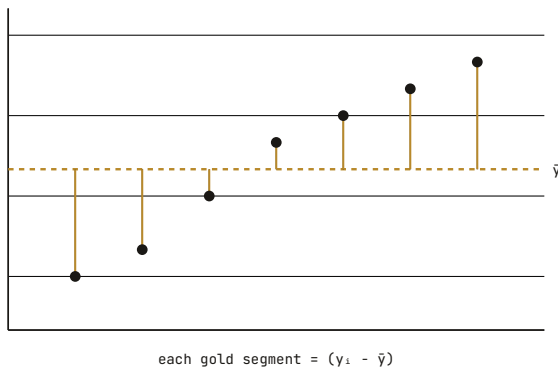
Residual (left over)

$$(y - \bar{y}) = (\hat{y} - \bar{y}) + (y - \hat{y})$$

Square each distance, sum across all points, and you get the three sums of squares.

01 *SST* — the total variation in y $\sum (y - \bar{y})^2$

SST (Sum of Squares Total) measures how much y jumps around its own mean. It has nothing to do with x or any model — it is simply the raw spread of the response variable. Think of it as the *budget* of variation you have to explain.



FORMULA

$$SST = \sum (y_i - \bar{y})^2$$

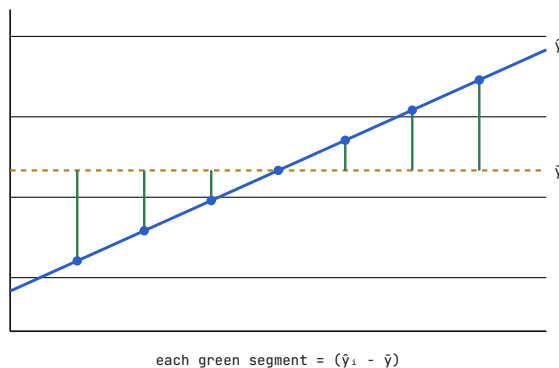
Imagine a world where you have no predictors — your best guess for every y is just the mean $\bar{y}$. SST is how wrong that guess is, squared and summed. It is the ceiling: **no model can explain more than SST.**

SST is fixed by your data. Fit any model you like — **SST does not change.**

02 **SSR** — what the model explains

$$\sum (\hat{y} - \bar{y})^2$$

SSR (Sum of Squares Regression) — sometimes called SSM or the "explained" sum of squares — measures how much of y 's variation the regression line accounts for. It is the distance from each prediction $\hat{y}$ to the mean $\bar{y}$, squared and summed.



FORMULA

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

A flat regression line would sit right on $\bar{y}$ and SSR would be zero — the model explains nothing. A steep line that tracks the data closely produces large $(\hat{y} - \bar{y})$ distances and a big SSR.

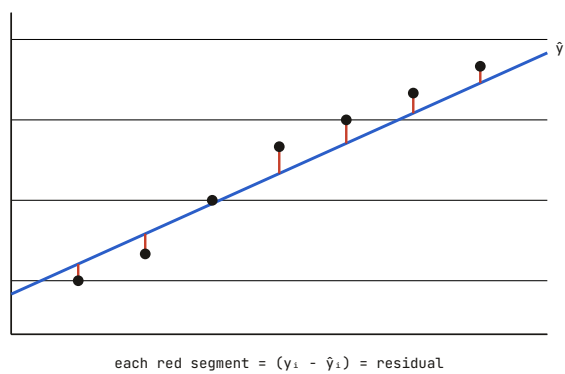
SSR = **variation the model captures**. Bigger SSR (relative to SST) means a better fit.

03 **SSE** — what the model misses

$$\sum (y - \hat{y})^2$$

SSE (Sum of Squares Error) — also called the residual sum of squares (RSS) —

is the leftover. The gap between each actual point and where the line *predicted* it would be. Ordinary least squares is literally the procedure that *minimizes* SSE.



FORMULA

$$SSE = \sum (y_i - \hat{y}_i)^2$$

Small SSE means the line hugs the data — the model is accurate. Large SSE means points are scattered far from predictions, and there is systematic structure the model failed to capture (or pure noise).

The fundamental identity: **SST**
= **SSR** + **SSE**. Budget =
Explained + Leftover.

EXAMPLE DECOMPOSITION • SST = 1,000

SST	total variation	1,000
SSR	explained by model	780
SSE	residual	220

04

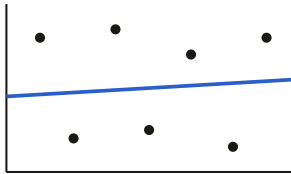
R^2 — the share explained

SSR / SST

R^2 (coefficient of determination) is simply the fraction of SST that the model explains. Equivalently, one minus the fraction it leaves unexplained. It lives between 0 and 1 — the higher, the better the fit.

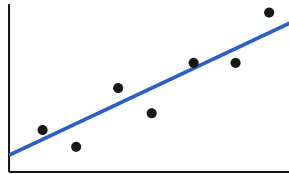
FORMULA

$$R^2 = SSR / SST = 1 - (SSE / SST)$$

LOW R^2 

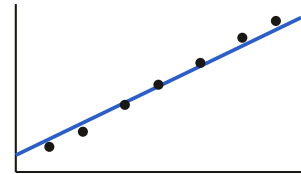
$$R^2 = 0.18$$

line barely helps

MEDIUM R^2 

$$R^2 = 0.64$$

trend + noise

HIGH R^2 

$$R^2 = 0.94$$

line hugs data

$R^2 = 0.78$ means **78% of the variation in y is explained by the model**; the remaining 22% is residual (unexplained by your predictors).

05 *MSR* & *MSE* — the mean squares

$$SS \div df$$

Sums of squares grow with sample size. To compare them fairly — and to build the F-test — we divide each by its **degrees of freedom**. The result is a *mean square*, which is basically an average squared deviation.

MEAN SQUARE REGRESSION

$$MSR = SSR / k$$

k = number of predictors

MEAN SQUARE ERROR

$$MSE = SSE / (n - k - 1)$$

n = sample size

MSR is the average amount of variation explained *per predictor*. If you add a useful predictor, MSR climbs.

MSE is the average squared residual — the estimated **variance of the errors** σ^2 . Its square root (RMSE) is in the original units of y and tells you typical prediction error.

$$F = MSR / MSE$$

The F-statistic for overall model significance is exactly this ratio — signal per predictor divided by noise per residual degree of freedom.

06 Adjusted R^2 — R^2 with a conscience

honest
fit

Here is the dirty secret of R^2 : **it never goes down when you add a predictor**, even a useless one. Throw in random noise as a variable and R^2 either stays flat or creeps up. That makes plain R^2 a terrible tool for comparing models of different sizes.

FORMULA

$$\text{Adjusted } R^2 = 1 - (1 - R^2) \times (n - 1) / (n - k - 1)$$

Adjusted R^2 penalizes you for each extra predictor. It only rises when a new variable helps *more than you'd expect by chance*. If you add junk, adjusted R^2 actually *falls*.

ADDING PREDICTORS TO A MODEL (N = 100)

MODEL	K PREDICTORS	R^2	ADJ. R^2	VERDICT
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Baseline	2 real	0.620	0.612	solid fit
+ useful predictor	3	0.710	0.701	both ↑ – keep it
+ weak predictor	4	0.714	0.702	$R^2 \uparrow$, Adj $\approx$ flat
+ random noise var	5	0.716	0.697	Adj. R^2 <i>dropped</i>

Rule of thumb: if you are choosing between models with different numbers of predictors, **trust adjusted R^2 over plain R^2 .**

SUMMARY

The whole family, at a *glance*

NAME	FORMULA	IN ONE SENTENCE
SST	$\Sigma(y - \bar{y})^2$	Total variation in y around its mean.
SSR	$\Sigma(\hat{y} - \bar{y})^2$	Variation explained by the regression line.
SSE	$\Sigma(y - \hat{y})^2$	Variation left unexplained – residuals squared.
MSR	SSR / k	Average explained variation per predictor.
MSE	SSE / (n-k-1)	Average residual variance – estimate of σ^2 .
R^2	SSR / SST	Share of variation the model explains.
Adj. R^2	$1 - (1-R^2) / (n-1)/(n-k-1)$	R^2 penalized for extra predictors.

END OF PRIMER



$$SST = SSR + SSE$$